

WHANHEE CHO

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RESEARCH INTERESTS

Data Analytics, Data Summarization, Applied LLMs

EDUCATION

UNIVERSITY OF UTAH, Graduate School, Utah, USA
Doctor of Philosophy in Computer Science

Aug. 2023 - Present

HANYANG UNIVERSITY, Seoul, Republic of Korea
Master of Science in Computer Science
Summa Cum Laude

Mar. 2021 - Feb. 2023

HANYANG UNIVERSITY, Seoul, Republic of Korea
Bachelor of Computer Science and Software Engineering
Summa Cum Laude

Mar. 2017 - Feb. 2021

PUBLICATIONS

- Aritra Mazumder, Whanhee Cho, Anna Fariha “QUWARTS: Query Workload Aware Relational Table Synthesis from Unstructured Text”, *VLDB 2026*,  **Best Demo Award Honorable Mention**
- Whanhee Cho, Shomit Fatin, Anna Fariha, “SAGE: Adaptive Recommendation of Spreadsheet Pivot Tables”, *SIGMOD, 2026*
- Whanhee Cho, Anna Fariha, “Data-Semantics-Aware Recommendation of Diverse Pivot Tables”, *SIGMOD, 2026*
- Whanhee Cho, Anna Fariha. “UTOPIA: Automatic Pivot Table Assistant”, *PVLDB, 2024*
- Minji Kim, Whanhee Cho, Soohyeong Kim, and Yong Suk Choi. “Simple Data Transformations for Mitigating the Syntactic Similarity to Improve Sentence Embeddings at Supervised Contrastive Learning”, *Advanced Intelligent Systems, 2024*
- Soohyeong Kim, Whanhee Cho, Minji Kim, and Yong Suk Choi. “Bidirectional Masked Self-attention and N-gram Span Attention for Constituency Parsing”, *EMNLP Findings, 2023*

PREPRINT

- Whanhee Cho, Kuangfei Long, Mahmood Jasim, Matteo Brucato, Alexandra Meliou, Peter J. Haas, Anna Fariha, “Example-Driven Intent Synthesis for Constrained Data Bundle Retrieval: Focused Text Snippet Extraction and Beyond”, 2026, (VLDB 2027 Under Revision)
- Wei-Xuan Wang, Whanhee Cho, Mahmood Jasim, Anna Fariha “CLUE: Answering How-To Questions on Dashboards using Metric Lineage Extractor and Simulator” 2026
- Whanhee Cho “ACCIO: Table Understanding Enhanced via Contrastive Learning with Aggregations.”, 2024

DATABASE MANAGEMENT RESEARCH EXPERIENCE

University of Utah DBLAB

Salt Lake City, USA

Research Assistant

Aug. 2023 - Present

- Intent-Aligned Interactive Bundle Retrieval (Under Revision at VLDB 2027) 
 - Developed a novel algorithmic framework for detecting multi-dimensional user intents for bundle retrieval without prompts, resolving combinatorial explosion via Integer Linear Programming (ILP) with PaQL and low computation cost. Demonstrated superior performance to RAGs on focused text snippet extraction tasks.
- Diversified Interesting Query Result Recommendation (Accepted at SIGMOD 2026) 
 - Designed a recommendation system that evaluates the usability of query results and leverages Large Language Models (LLMs) to capture human common sense, successfully increasing the surprisingness and utility of the output.
 - Solved the combinatorial explosion of candidate sets with pruning strategies, reducing candidate evaluation cost by 80% while maintaining quality. We evaluated our work and novel metrics via a formal user study.
- Semantic aware data post-processing in the pivot table (Published at VLDB 2024) 

- Designed a system that leverages language models and clustering algorithms to automatically group related entities and correct typos, delivering semantically accurate pivot tables in spreadsheet.
- Table Understanding with Contrastive Learning
 - Introduced novel contrastive learning paradigm using related table pairs, achieving 91.1% micro-F1 on column type annotation. Demonstrated competitive performance approaching state-of-the-art methods (within 4% on micro-F1) while using conceptually simpler approach without data augmentation.

AI RESEARCH EXPERIENCE

Hanyang University Artificial Intelligence LAB

Seoul, South Korea

Full-time Student Researcher

Dec. 2020 - Feb. 2023

- Proposed and implemented a method to syntactically augment a dataset to improve sentence embedding performance. Our work demonstrates that simple transformations can have a significant impact on model performance, addressing the limitations of existing semantic-only approaches
- Leverage the prior works' exploration that the internal layers of language encoders learn different aspects of a sentence. We demonstrated that using these varied outputs as input for a GAN produces a more meaningful representation than using random noise, which often fails in low-resource settings.
- Data preprocessing, and data filtering for pre-training large-scale A.I. model of KT AI One Team*. Led the development of a toxic dataset and classifier for pre-training large-scale AI models, approaching data filtering as a classification task to protect the model from training with toxic data. (KT AI One Team*: A national AI cooperative organization of 12 industry-university-research institutes.)
- Investigated relations and impacts of next token probability onto positional embeddings. The purpose of this project was to validate the effect of GPT-2 next token prediction probability on positional embedding at the relation extraction task using the TACRED dataset. According to the next token probability, set the positional embedding values of the next gold token to be closer or further from the current token.

TECHNICAL SKILLS

- Programming Languages: C/C++, Python, Java, SQL
- Tools & Technologies: Git, CMake, Linux kernel programming, CUDA, Docker
- LLMs/ML: Huggingface, vLLM, PyTorch, TensorFlow, scikit-learn

CONFERENCE & PRESENTATION

- SIGMOD & HILDA, "Data-Semantics-Aware Recommendation of Diverse Pivot Tables", Bengaluru, India, 2026
- University of Utah, School of Computing, "Data Science Research," Salt Lake City, USA, April 2025
- VLDB, "UTOPIA: Automatic Pivot Table Assistant", Guangzhou, China, August 2024
- NWDS, "UTOPIA: Automatic Pivot Table Assistant," Seattle, USA, August 2023

TEACHING EXPERIENCE

Teaching Assistant | University of Utah

- CS6959/3960: Human-Centered Data Management | Spring 2026
- CS6530: Advanced Database Systems | Fall 2025

Teaching Assistant | Hanyang University

- ITE4053: Deep Learning Methods and Applications | Fall 2022
- CSE4007: Artificial Intelligence | Fall 2021, Spring 2022

AWARDS & HONORS

SIGMOD Student Travel Grant	2026
Fellowship, Kahlert School of Computing, University of Utah	2023
Brain Korea 21 Fellowship, NRF Korea	2021-2022
Need Based Scholarship, Hanyang University	2019-2020
Merit Based Scholarship, Hanyang University	2018, 2021-2022
Honors student in the 1st semester of 2018, Hanyang University	2018
National Grant, Korea Student Aid Foundation (KOSAF)	2017- 2020